

## Reference Guide for the Course

### 1. R Packages

**dplyr:** For manipulating data. <https://www.rdocumentation.org/packages/dplyr/versions/1.0.10>

**ggplot2:** For data visualization. <https://ggplot2.tidyverse.org/>

**naniar:** For missing data assessment. <https://naniar.njtierney.com/>

**stringr:** For manipulating string data. <https://stringr.tidyverse.org/>

**lubridate:** For manipulating date data. <https://lubridate.tidyverse.org/>

**janitor:** For cleaning data. <https://www.rdocumentation.org/packages/janitor/versions/2.2.1>

**Epi:** Summarizes regression outputs. <https://www.rdocumentation.org/packages/Epi/versions/2.63>

**car:** Assesses regression diagnostics. <https://www.rdocumentation.org/packages/car/versions/3.1-5>

**visdat:** Assist ggplot2 package for data visualization.  
<https://www.rdocumentation.org/packages/visdat/versions/0.6.0>

**pmsampsize:** Calculates sample sizes for prediction model development. <https://cran.r-project.org/web/packages/pmsampsize/index.html>

**pROC:** Calculates area under the curve for predictions.  
<https://www.rdocumentation.org/packages/pROC/versions/1.19.0.1>

**CalibrationCurves:** Visualizes calibration plots for predictions. <https://cran.r-project.org/web/packages/CalibrationCurves/index.html>

**Hmisc:** Helper package for prediction (will not use any direct functions):  
<https://www.rdocumentation.org/packages/Hmisc/versions/5.2-4>

**rms:** Helper package for prediction (will not use any direct functions):  
<https://www.rdocumentation.org/packages/rms/versions/8.1-1>

## 2. Pre-Course

**Downloading and opening R and R Studio:**

<https://www.r-project.org/>

<https://posit.co/download/rstudio-desktop/>

**Installing R Tools:**

<https://cran.r-project.org/bin/windows/Rtools/>

**install.packages() :**

R function to install packages.

**Downloading data from Kaggle:**

<https://www.kaggle.com/datasets>.

**read.csv(filepath) :**

R function to upload data to R.

**library(RPackage) :**

Activate a library to use in R

**%>% :**

‘Pipe’ to connect multiple functions

**Tidy data:**

Standardized way to order data, where each column is a variable and each row is an observation

### 3. Open Science and Data Protection

#### **FAIR Principles for Open Science**

*F — Findable*

Data and metadata should be easy for both humans and computers to discover.

*A — Accessible*

Once found, data can be retrieved using well-defined procedures.

*I — Interoperable*

Data can be combined with other data and used in different systems.

*R — Reusable*

Data can be reused for future research and new purposes.

**De-Identification:** Basic removing names and birthdates. Creating a unique ID

**Pseudonymization:** Renaming patients by an independent third party to improve patient data protection.

**Data Caretaker:** “Gate Keeper.” Limited disclosure of personal health information under a data caretaker that limits the scope which the data may be used or released.

**Federated Data System:** Database systems that are autonomous, where data are not replicated or copied from the original database.

**Data Dictionary:** Dictionary of each variable in a dataset, including definition, units, and any transformations performed.

**Four eyes principle:** All code should be checked by at least one other person ‘four eyes’ before depositing code.

**set.seed()** : ‘Setting seed’ functions allows for reproducibility of a randomized process.

**anonymize( , .algo = "crc32")** : Creates a random ID. Anonymizer package.

**sample(variable, #, replace = FALSE)** : Function that randomly samples from a variable. Can detail how many samples (#), and if you do or do not want to have replacement for sampling.

#### 4. Joining & Transforming Data

**Wide Data:** Each subject has one row. Different measurement time points or conditions are separate columns

**Long Data:** Each row is a single observation. Each time point or condition is a different row

**filter()** : function: changes which rows are present without changing their order. Dplyr package.

**arrange()** : changes the order of the rows without changing which are present. Dplyr package.

**distinct()** : gives on exact match per row. Dplyr package.

**%in%** : Keeps rows where the variable equals one of the values on the right. Dplyr package.

**mutate()** : Creates new columns from other columns. Dplyr package.

**select()** : Changes which columns are present. Dplyr package.

**rename()** : Changes the names of the columns. Dplyr package.

**relocate()** : Changes the positions of the columns. Dplyr package.

**clean\_names()** : Lower caps and cleans all variable names to computer readable format. Janitor package.

**group\_by()** : Groups by different logic (left v right; male v female). Dplyr package.

**summarize()** : Aggregates by different commands (mean, median, etc.). Dplyr package.

**slice()** : Extract different rows from each group. Dplyr package.

**left\_join()** : The data frame on the left will have the same # rows. Dplyr package.

**inner\_join()** : Will only join data that is in BOTH. Dplyr package.

**right\_join()** : The data frame on the right will have the same # rows. Dplyr package.

**full\_join()** : All rows from BOTH data frames are included. Dplyr package.

**semi\_join()** : Keeps only rows that are in the left with a match in right. Dplyr package.

**anti\_join()** : Row that are NOT in BOTH data frames are included. Dplyr package.

## 5. Cleaning Data

**View()** : Views the entire dataset in tabular 'Excel' format. Dplyr package.

**head()** : Provides the first 10 rows of the dataset. Dplyr package.

**tail()** : Provides the last 10 rows of the dataset. Dplyr package.

**names()** : Provides a list of every variable name within the dataset. Base R.

**str()** : Provides the structure of each variables within the dataset. Base R.

**hist()** : Creates a histogram of a variable. Base R.

**count()** : Counts the number of observations of a variable. Dplyr package.

**str\_trim** : Trims the spaces on both side of a string that are 'empty'. Stringr package.

## 6. Data Visualization

All of these functions are from the ggplot2 package.

**ggplot(, aes( ))** : Base ggplot2 function, includes dataframe, and X and Y at a minimum.

+ **geom\_point()** : Addition to ggplot function, to create a point for each piece of data.

+ **geom\_smooth()** : Addition to ggplot function, to create a smooth line for the data.

+ **xlab()** : Addition to ggplot function, x-axis lab title function.

+ **ylab()** : Addition to ggplot function, y-axis lab title function.

+ **ggtitle()** : Addition to ggplot function, overall figure title function.

+ **labs()** : Addition to ggplot function, legend title function.

+ **scale\_color\_discrete()** : Addition to ggplot function, name each color level within the legend.

+ **geom\_jitter()** : Addition to ggplot function, 'jitters' data points.

+ **facet\_wrap()** : Addition to ggplot function, creates equal units figure side by side.

## 7. Missing Data

**Missing Completely at Random (MCAR):** Data is missing without a pattern.

**Missing at Random (MAR):** Data is missing with a pattern, but it is explained by the data at hand within the dataset.

**Missing Not at Random (MNAR):** Data is missing with a pattern, but is cannot be explained by the data at hand within he dataset.

**sum(is.na())** : Base R function that gives the total number of missing data points within a dataset. Base R.

**dim()** : Base R function that provides the dimensions of a data set columns and rows. Base R.

**miss\_var\_summary()** : Provides the number and percent missing per variable in a dataset. Naniar package.

**vis\_dat()** : Provides missing visuals of all data points within a dataset. Naniar package.

**geom\_miss\_point()** : Part of ggplot2 function, highlights missing data points. Ggplot2 & Naniar packages.

**vis\_miss()** : Provides visual of missing data points by cluster. Naniar package.

**gg\_miss\_upset()** : Shows patterns of missingness by variable clusters. Ggplot2 & Naniar Packages.

**gg\_miss\_var()** : Shows cumulative missingness by variable. Ggplot2 & Naniar Packages.

**bind\_shadow()** : Part of the shadow matrix, which provides each missing point as as new data point to compare against data that is observed. Naniar package.

## 8. Epidemiology

**Epidemiology:** The scientific study of the distribution, patterns, and determinants of health-related states or events (including diseases) in specified populations.

**Exposure:** Time at risk.

**Injury:** Physical damage to the body's tissues as a consequence of energy transfer and develops over a short period of time.

**Recurrent Injury:**

**Subsequent Injury:**

**International Classification of Diseases:** Classification of diseases by the World Health Organization

**Occupational Injury & Illness Classification:** Classification of injury and illness by the US Dept of Labor.

**The Orchard Sports Injury Classification System (OSIICS):** Classification of injury by body part for sport.

**PICO:** Population; Intervention; Comparison; Outcome.

**Confounding:** A systematic error in epidemiology where a third variable (confounder) distorts the true relationship between an exposure and an outcome, causing an apparent, yet false, association.

**Person Time:** The number of exposures for all involved patients.

**Risk:** Average probability of incurring the disease per patient.

**Censor:** Patient occurrence of disease does not happen during observation.

**Right censor:** Follow up ends without disease.

**Left Censor:** Disease happens before observation.

**Incidence:** The number of cases of disease with their onset during a prescribed period of time.

**Rate:** The number of cases of disease with their onset per unit of person time.

**Prevalence:** The number of cases with a disease or attribute (i.e., injury) in a specific population.

**min()** : Function that gives the minimum number in a variable. Base R.

**max()** : Function that gives the maximum number in a variable. Base R.

**sum()** : Function that sums the total numbers in a variable. Base R.



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**round()** : Function that rounds a number to prescribed number of digits. Base R.

## 9. Confounding

**Confounding:** Alternative more lay definition: The relationship between an independent variable (exposure) and a dependent variable (outcome) is distorted by the presence of another variable.

**Association:** Describe.

**Predict:** Forecast into the future.

**Causal:** Cause-effect.

**Three Stages of Identifying Confounding:** 1) Univariable Analyses; 2) Add Potential Covariates; 3) Externally Replicate and/or Evaluate in Different Scenarios.

**Creating Confounding Structural Relationships:** 1) Search & synthesize the literature; 2) Evaluate biological plausibility; 3) Interview experts (if necessary); 4) Create a structural diagram to explicitly detail this relationship.

**Diagonal Acyclic Graph (DAG):** Visually representation of causal relationships.

- **Directed:** The edges (arrows) between nodes (variables) have a direction, indicating a causal or dependency relationship
- **Acyclic:** There are no loops or cycles — you cannot start at one node and follow a path that leads back to it.

**Common Cause:** To both the exposure & outcome.

**Collider:** Caused by two or more other variables. Conditioning on the collider introduces bias into the analyses.

**Mediator:** A part of the causal pathway. Is caused by the exposure and influences the outcome. Transmits part or all of the effect of the exposure to the outcome.

## 10. Regression

**Continuous Data:** Quantitative variables that can any value within a given range, including decimal or fractions. It is measured, not counted.

**Binary Data:** A categorical variable that can only take two possible values, usually yes/no, true/false, or 0/1.

**Ordinal Data:** A categorical variable where the categories have a meaningful order/ranking, but the differences between categories are not numerically measured.

**Count Data:** Numerical data that represents a number of times an event occurs. Non-negative whole numbers and cannot be a fraction or decimal.

### Basic Steps in A Regression Analysis:

- Identify your question to be answered
- Understand your data
- Identify the right analytical strategy
- Identify potential covariates (i.e., confounders) to include in the analysis
- Perform regression
- Assess assumptions through diagnostics
- Iterate until the analyses meet the assumption
- Infer conclusions

**P-Value:** Probability of observing a result greater than this.

**95% Confidence Interval:** If the analysis was performed 100 times you would expect the results to be within this bound 95 times.

**Linear Regression:** Regression system used for a continuous data outcome.

**Logistic Regression:** Regression system used for a binary outcome.

**Poisson Regression:** Regression system used for a count outcome.

**Ordinal Regression:** Regression system used for an ordinal Outcome.

**Intercept:** Expected mean value of the outcome (Y), when all exposures and covariates are held at 0.

**Coefficient:** Estimated change in the outcome (Y) with a one unit increase in the exposure/covariate (X).

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**Error:** The unobserved variable, determined as the difference in actual versus estimated data.

**Residuals:** Difference of actual versus estimated regression points. Used for diagnostics of the regression model.

**Assumptions:** Diagnostic markers that need to be held to provide valid and reliable results from a regression. Assumptions are based off of the residuals

- Linearity
- Independence of each observation
- Normality
- Constant (equal) variance

**Homoscedasticity:** Constant variance between each point in a regression residuals.

**Heteroscedasticity:** Non-constant variance between each point in a regression residuals. This violates regression assumptions.

**Leverage:** The 'weight' of one point and how this effects the overall regression.

**Outlier:** An extreme value that is beyond a particular spread of the data. These data points can overly affect regression model results.

**Beta:** Coefficient for a linear regression model. 0 is the null.

**Odd Ratio:** Coefficient for a logistic regression model. 1 is the null.

**Multicollinearity:** When two or more variables are highly correlated.

**Variance Inflation Factor (VIF):** Measures severity of correlation. How much variance inflation is due to correlation

**glm(Y ~ X, data = data, family = 'gaussian') :** Basic regression model formula in R. Base R.

**plot(model) :** Function to assess regression model diagnostics. Base R.

**summary( ) :** Summarizes the basic regression model results. Base R.

**ci.lin(model) :** Summarizes the basic regression model results with 95% confidence intervals. Epi package.

**ci.lin(model, Exp = T) :** Summarizes the basic regression model results with 95% confidence intervals, to obtain Odds Ratios from a logistic regression. Epi package.

**vif(model) :** Assesses variance inflation of the model. Car package.

**log() :** Performs logarithmic transformation of a variable. Base R.

## 11. Prediction

**Prediction:** The process of developing and testing mathematical models (equations) that include multiple factors (obtained from descriptive and causal research) that forecast the risk/probability of an undiagnosed condition (diagnostic) or future outcome (prognostic).

### Basic steps in a prediction model:

- Define outcome
- Identify potential data collection method and/or data sets
- Determine sample size
- Assess missing data
- Predictor selection
- Assess potential for predictor non-linearity
- Model development
- Internal validation Prediction performance assessment
- External and transportation validation

**Parameter:** The number of independent pieces of information in a model that are used to estimate and/or forecast another piece of information (i.e., prediction risk score or outcome).

**Predictor:** A variable that helps predict/forecast an outcome.

**Overfitting:** When a prediction model learns the intricacies and patterns in a data set too well, and is less generalizable to external data.

**Regularization:** Method to prevent overfitting by adding a penalty term to the model's loss function.

**Optimism Bias:** Developed data will perform better than external data.

**Internal Validation:** Assessment of prediction model through using the same data multiple times either through bootstrapping or 10-fold cross-validation.

**Bootstrapping:** Repeatedly resampling a single original dataset with replacement to create multiple "bootstrap samples".

**10-fold cross-validation:** Partitioning data into 10 equal subsets ("folds"). The model is trained 10 times, each time using 9 folds (90%) for training and 1 fold (10%) for validation.

**Internal-External Cross-Validation:** Internal validation followed by splitting data by cluster to further test the stability of the prediction performance.

**Discrimination:** How well a model differentiate between individuals with the outcome and individuals without it. Tested through Area Under the Curve (AUC) or c-statistic.

**Calibration:** The agreement between predicted risks (derived from a model) and observed risks in the data set. Tested through slope with 95% confidence intervals and visually.

**R<sup>2</sup>:** Model explanation of overall variance.

**Calibration in the Large:** How close a model's average predicted risk is to the overall proportion with the event in the data set. This indicates whether the predictions are systematically too low or too high.

**TRIPOD AI+:** Reporting guideline designed for prediction. <https://www.equator-network.org/reporting-guidelines/tripod-statement/>

**External Validation:** Testing the prediction algorithm on a new dataset.

**Temporal Validation:** Testing the prediction model on data from a new time period

**Transportation Validation:** Testing the prediction model on data from a different country or population.

**pmsampsize()** : Function to calculate sample size for prediction models. Pmsampsize R package.

**roc(y ~ prediction, ci=TRUE)** : Area under the curve function. pROC package.

**offset()** : Function that creates an offset in a model equation. Used in this case to help calculate the calibration in the large. Base R.

**mean ()** : Calculates mean. Base R.

**quantile()** : Calculates defined quantile. Base R.

**colnames()** : Defines column names. Base R.

**r\_square <- 1 - model\$deviance/model\$null.deviance** : Function to calculate R<sup>2</sup>. Base R

**roc()** : Function to calculate area under the curve. pROC R Package.

**val.prob.ci.2()** : Function to visualize calibration slope from predictions. CalibrationCurves R Package.